0.0.1 Locally Compact

Definition 0.0.1. A Space X is called *Locally Compact* if:

For any $x \in X$, there exist open U and compact C such that $x \in U \subseteq C$.

0.0.2 One-point Compactification

Definition 0.0.2. Let $(X, \mathcal{T})$ be a Space.
Define $X_\infty \overset{\text{def}}{=} X \sqcup \{\infty\}$ and $\mathcal{T}_\infty \overset{\text{def}}{=} \mathcal{T} \sqcup \{U \subseteq X_\infty \mid \infty \in U,\ X_\infty \setminus U \text{ is compact in } X\}$.
This $(X_\infty, \mathcal{T}_\infty)$ is called *one-point compactification* of X.

Theorem 0.0.3. Let (X, ∞) be a Locally-Compact Hausdorff Space, but not Compact.
Then, one-point compactification $(X_\infty, \mathcal{T}_\infty)$ of X is Compact Hausdorff Space.

Proof. This proof consisted of five steps.
1). Claim: $\mathcal{T}_\infty$ is Topology on X_∞. (Using X is Hausdorff)
Let $U_\gamma \in \Gamma,\ (\gamma \in \Gamma)$ be elements of $\mathcal{T}_\infty$.
Define $\Gamma_1 \overset{\text{def}}{=} \{\alpha \in \Gamma \mid U_\alpha \in \mathcal{T}\}$, and $\Gamma_2 \overset{\text{def}}{=} \Gamma \setminus \Gamma_1 = \{\beta \in \Gamma \mid \infty \in U,\ X_\infty \setminus U \text{ is compact in } X\}$.
Then, $\displaystyle\bigcup_{\gamma \in \Gamma} U_\gamma = \left(\bigcup_{\alpha \in \Gamma_1} U_\alpha\right) \cup \left(\bigcup_{\beta \in \Gamma_2} U_\beta\right)$. The left term is open in X clearly.
And, put $C_\beta = X_\infty \setminus U_\beta$ for each $\beta \in \Gamma_2$. Then, C_β is Compact in X by definition, thus closed by X is Hausdorff.

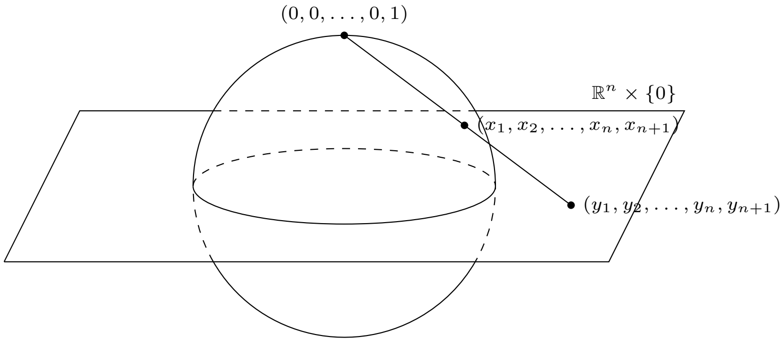
$$\bigcup_{\beta \in \Gamma_2} U_\beta = \bigcup_{\beta \in \Gamma_2} X_\infty \setminus C_\beta = X_\infty \setminus \left(\bigcap_{\beta \in \Gamma_2} C_\beta\right)$$

This intersection of C_β is compact, being any intersection of closed is closed and closed subset of compact.
That is, it is compact in X, therefore this union of U_β is contained in $\mathcal{T}_\infty$.
Let $U_1, U_2 \in \mathcal{T}$, and $V_1, V_2 \in \mathcal{T}_\infty \setminus \mathcal{T}$. Put $C_i \overset{\text{def}}{=} X_\infty \setminus V_i,\ (i = 1, 2)$. Then, C_i is compact. Now,

$$\begin{aligned}
&U_1 \cap U_2 \in \mathcal{T} \subset \mathcal{T}_\infty \\
&U_1 \cap V_1 = U_1 \cap (X_\infty \setminus C_1) = U_1 \cap X_\infty \cap C_1^c = U_1 \cap C_1^c = U_1 \setminus C_1 \in \mathcal{T} \subset \mathcal{T}_\infty \\
&V_1 \cap V_2 = (X_\infty \setminus C_1) \cap (X_\infty \setminus C_2) = X_\infty \setminus (C_1 \cap C_2) \in \mathcal{T}_\infty
\end{aligned}$$

Thus closed under the arbitrary union and finite intersection.
2). Claim: $(X, \mathcal{T})$ is a Subspace of $(X_\infty, \mathcal{T}_\infty)$. That is, $\mathcal{T} = \{U \cap X \mid U \in \mathcal{T}_\infty\}$. (Using X is Hausdorff)
The right inclusion is clear: $U \in \mathcal{T} \implies U \in \mathcal{T}_\infty$. Thus $U = X \cap U \in \{U \cap X \mid U \in \mathcal{T}_\infty\}$.
To show the left inclusion: Let $U \in \mathcal{T}_\infty$. If $U \in \mathcal{T}$, then $X \cap U = U \in \mathcal{T}$.
If $U \notin \mathcal{T}$, then $X_\infty \setminus U$ is compact in X. Now, $X \cap U = X \setminus \underbrace{(X_\infty \setminus U)}_{\text{compact in } T_2 \implies \text{closed}} \in \mathcal{T}$.
3). Claim: $\overline{X} = X_\infty$. That is, closure of X is X_∞. (Using X is not compact)
Let $U \in \mathcal{T}_\infty$ with $\infty \in U$. Then, $X_\infty \setminus U$ is compact of X, thus $X_\infty \setminus U \subsetneq X$ because X is not compact.
4). Claim: X_∞ is Compact Space.
Let $\mathcal{O} = \{U_\alpha \mid \alpha \in \Lambda\}$ be an open cover of X_∞. Since $\infty \in X_\infty = \displaystyle\bigcup_{\alpha \in \Lambda} U_\alpha$, there is $\alpha_0 \in \Lambda$ such that $\infty \in U_{\alpha_0}$.
$C \overset{\text{def}}{=} X_\infty \setminus U_{\alpha_0}$ is compact in X, thus so in X_∞. And, $C \subseteq \displaystyle\bigcup_{\alpha \in \Lambda \setminus \{\alpha_0\}} U_\alpha$, thus there is finite subcover of C.
Finally, union of finite subcover of C and U_{α_0} is finite subcover of X_∞.
5). Claim: X_∞ is Hausdorff. (Using X is Locally-Compact)
Let $x, y \in X_\infty$. If both x, y are contained X, then there is nothing to prove, being X is hausdorff.
If $x \in X$ and $y = \infty$, then there is open U and compact C of X such that $x \in U \subseteq C$, by Locaaly-Compact.
Now, $x \in U$ and $\infty \in X_\infty \setminus C$, both are open of X_∞ with $U \cap (X_\infty \setminus C) = \emptyset$. $\square$

0.0.3 Stereographic projection



Definition 0.0.4. Let $\mathbb{R}^{n+1}$ be a Euclidean Space. Define *Unit Sphere* on $\mathbb{R}^{n+1}$:

$$S^n \overset{\text{def}}{=} \{(x_1, x_2, \dots, x_n, x_{n+1}) \in \mathbb{R}^{n+1} \mid \|(x_1, \dots, x_{n+1})\| = 1\}$$

Theorem 0.0.5. For any $p \in S^n$, $S^n \setminus \{p\} \cong \mathbb{R}^n$.

Proof. WLOG, put $p = (0, 0, \dots, 0, 1)$. Define a map *Stereographic Projection*:

$$f : S^n \setminus \{p\} \to \mathbb{R}^n : (x_1, x_2, \dots, x_n, x_{n+1}) \mapsto \left(\frac{x_1}{1 - x_{n+1}}, \frac{x_2}{1 - x_{n+1}}, \cdots, \frac{x_n}{1 - x_{n+1}}\right)$$

This map is derived by:

$$(y_1, y_2, \dots, y_n, y_{n+1}) = t(\overrightarrow{x} - \overrightarrow{p}) + (0, 0, \dots, 0, 1) = (x_1 t, x_2 t, \dots, x_n t, (x_{n+1} - 1)t + 1)$$

$y_{n+1} = 0$ when $t = \dfrac{1}{1 - x_{n+1}}$. Each $(x_i, x_{n+1}) \mapsto \dfrac{x_i}{1 - x_{n+1}}$ is Continuous map, so f is Countinuous map. And, the map

$$g : \mathbb{R}^n \to S^n \setminus \{p\} : (x_i)_{i=1}^n \mapsto \left(\frac{2x_1}{1 + \|(x_i)\|^2}, \frac{2x_2}{1 + \|(x_i)\|^2}, \cdots, \frac{2x_n}{1 + \|(x_i)\|^2}, 1 - \frac{2}{1 + \|(x_i)\|^2}\right)$$

is Continuous, and $g = f^{-1}$. Thus, f is Homeomorphism. □

Theorem 0.0.6. One Point Compactification of $\mathbb{R}^n$ is $S^n \subseteq \mathbb{R}^{n+1}$.

Proof. Since $\mathbb{R}$ is Locally-Compact Hausdorff Space, finite product Space $\mathbb{R}^n$ is Locally-Compact Hausdroff Space. Let $\mathbb{R}^n_\infty = \mathbb{R}^n \cup \{\infty\}$ be a One Point Compactification of $\mathbb{R}^n$. Then, $\mathbb{R}^n_\infty$ is Compact Hausdorff Space. Meanwhile, put $p = (0, 0, \dots, 0, 1)$. Since $S^n \setminus \{p\} \cong \mathbb{R}^n$, there exists a Homeomorphism $f : S^n \setminus \{p\} \to \mathbb{R}$. Define

$$\tilde{f} : S^n \to \mathbb{R}^n_\infty : \begin{cases} f(x) & x \neq p \\ \infty & x = p \end{cases}$$

Then, $\tilde{f}$ is Bijective map, and Continuous because: for any open $U \subseteq \mathbb{R}^n_\infty$ containing ∞,

$$\tilde{f}^{-1}[\mathbb{R}^n_\infty \setminus U] = f^{-1}[\mathbb{R}^n \setminus U]$$

Since $\mathbb{R}^n \setminus U$ is Compact and f^{-1} is continuous, $\tilde{f}^{-1}[\mathbb{R}^n_\infty \setminus U] = f^{-1}[\mathbb{R}^n \setminus U]$ is compact in S^n, thus closed. Moreover, since S^n is Compact and $\mathbb{R}^n_\infty$ is Hausdorff, $\tilde{f}$ is Closed map. Consequently, $\tilde{f}$ is Homeomorphism. □